

working at high frequencies in the olden days.

When active filters were used, built using vacuum tubes or transistors, they generally had higher gains at lower frequencies. So fewer amplifier stages were required if the frequencies were lower due to high gains at the lower frequencies. Thus, the choice of having IF above the AM band (i.e., above 1602kHz) was ruled out.

3. Be as high as possible. The higher IF provides a sufficient bandwidth to accommodate the audio signals without distortion, ensuring clear and high-quality sound reproduction. Of course, it should still be lower than the lowest of the AM band (531kHz).

Further, the higher the IF, higher the gap between tuned frequency ($f_2 - f_1$) and the image frequency. This higher gap helps in filtering out the image frequency more easily. (We will discuss the image frequency soon.)

4. Not be matched by the difference of any two channels. Two strong stations whose frequencies happen to be separated by the IF could produce IF. This unwanted IF generated from such interference will get through filters, which are designed to allow IF to the next stage, such as detector. In other words, the IF should not be 10kHz or 9kHz, or even the multiples of 10kHz or 9kHz. That means any number that is divisible by 10 or 9 is ruled out.

Taking all these into consideration, many frequencies were popular

Typical IF values

The value of the intermediate frequency varies based on the application. For example, in standard AM radios, it's typically around 455kHz, while in FM radios, it's around 10.7MHz. In television receivers, the IF might be around 38MHz for video and 33.4MHz for audio.

What is Heterodyning

The process of mixing two signals, i.e., modulated carrier wave (f_1) and the signal generated by the local oscillator (f_2), is called heterodyning. Also called frequency conversion and beat frequency generation, it is very widely used in communications engineering to generate new frequencies and move information from one frequency channel to another.

The resulting output of the mixer is two signals: addition and subtraction of the frequencies of the two input signals (f_1 and f_2), one the sum of the two frequencies ($f_1 + f_2$), and the other the difference of the two frequencies ($f_1 - f_2$). The new signal frequencies are called 'heterodynes.' Typically, only one of the heterodynes is required and the other signal is filtered out.

The word heterodyne is coined from the Greek language 'hetero' (different) and 'dyn' (power). The heterodyning was first developed in early 1900's.

Super heterodyning or superhet. It is basically same as heterodyne, but all the signals involved, more particularly the IF, are beyond the audible range to prevent any interference with the audio signal (higher than 20,000Hz or 20kHz), hence the name 'supersonic heterodyning', which is often referred as 'super heterodyning' or 'superhet.'

choices during World War I time-frame, in early 1900's.

As we can see, dealing with higher frequencies, even in the range of a few kilohertz (kHz), was challenging in those early days due to unavailability of suitable components. This resulted in radio makers working with lower frequencies for IF. However, this resulted in poor audio quality when the audio was extracted from very low IF, such as 30 or 35kHz. Then the competition began to build better systems with higher IF to generate better audio qualities in radios.

Many radio manufacturers started making radios in those days with higher and higher IF, such as 100, 155, 300, 455, 500, or even 700kHz! While higher IF produced better audio quality, it obviously resulted in a higher cost.

After World War I came the Great Depression (1929-1939), just before the World War II. It was a period of economic depression in the United States. Naturally, demand for low-cost systems, components, and interchangeable parts in the radios went up. The 455kHz appeared like a good choice considering the good-quality audio it could produce and the lower cost for components

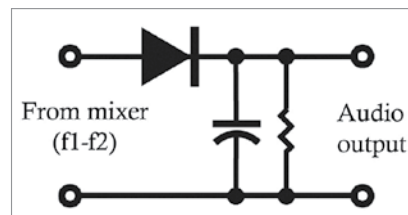


Fig. 4: A simple envelope detector (also known as audio extractor, audio detector, or demodulator)

compared to other higher IF choices. So, then 455kHz was adopted as a standard by the industry.

Image frequency

Let us quickly understand what image frequency is, and why and how it should be eliminated.

Let us say an AM radio receiver's IF is 455kHz. Suppose we want to tune to a radio station broadcasting at 610kHz. That means we need to get the local oscillator (LO) to produce 1065kHz so that when we send LO output of 1065kHz and broadcast frequency of 610kHz to a mixer, the mixer produces the difference of 1065kHz and 610kHz as 455kHz, which is our desired IF.

We have a problem here.

If there is another radio station broadcasting at 1520kHz, then the same mixer with its local oscillator supplying 1065kHz will produce the